

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA0502	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO5 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)</i>		
Student Name:	Michelle Joanna G	Reg. No.:	192325005
Section / Batch:	SLOT C / 2026	Date:	25/07/2026

Code of Conduct

*I **Michelle Joanna G(192325005)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Michelle Joanna G

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
GRAND TOTAL:			20 Marks

Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

- (a) Interpret the scenario and identify the type of noise present.**
- (b) Explain how this degradation affects inspection accuracy.**
- (c) Recommend an appropriate enhancement technique to remove the interference.**
- (d) Justify why your selected approach is more suitable than spatial-domain methods.**
- (e) Explain your answer in a logical sequence.**

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

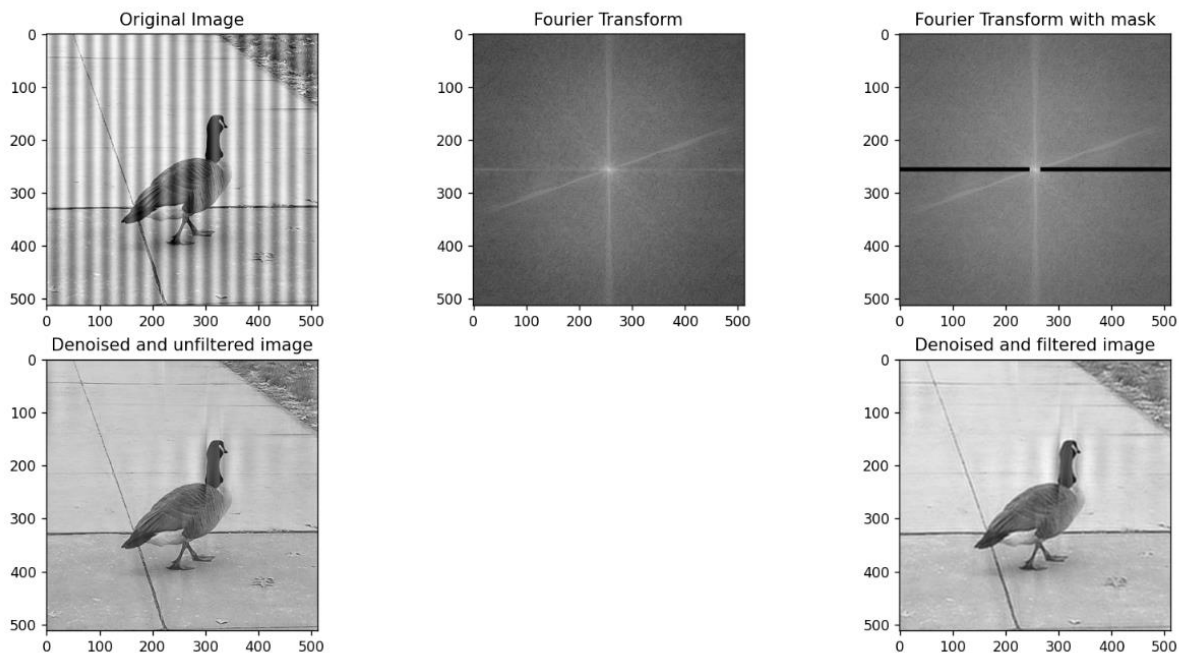
- (a) Interpret the scenario and explain the challenge involved.**
- (b) Identify the importance of preserving edges while removing noise.**
- (c) Recommend a suitable filtering technique.**
- (d) Justify why your selected filter preserves important image details.**
- (e) Present your explanation clearly and systematically.**

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)		Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Scenario 1: Manufacturing Quality Inspection



(a) Interpret the scenario and identify the type of noise present.

The automated inspection system captures images of metal components for defect detection. Due to electrical interference during image acquisition, repetitive horizontal stripes appear across the images.

Type of noise:

- **Periodic noise (stripe noise)** caused by electrical interference.
- It appears as regularly spaced horizontal lines and is most easily identified in the frequency domain.

(b) Explain how this degradation affects inspection accuracy.

The horizontal stripes reduce image quality by covering important surface details.

Effects include:

- Small cracks and scratches may become hidden.
- False defects may be detected because the stripes resemble defects.
- Edge detection becomes less accurate.
- Automated quality inspection systems may produce incorrect classifications, reducing production reliability.

(c) Recommend an appropriate enhancement technique to remove the interference.

Recommended Technique:

Frequency Domain Filtering using a Notch Reject Filter

Working:

Recommended technique:

- Frequency-domain filtering using the Fourier Transform (FFT) with a Notch Filter (Band-Reject Filter).

Process:

1. Convert the image into the frequency domain using FFT.
2. Locate the bright frequency peaks caused by the periodic stripes.
3. Apply a notch filter to remove only those frequencies.
4. Perform the inverse FFT to reconstruct the cleaned image.

(d) Justify why your selected approach is more suitable than spatial-domain methods.

A **Notch Reject Filter** is more suitable because:

Frequency Domain (Notch Filter)	Spatial Domain Filters
Removes only periodic noise frequencies	Removes noise from the whole image
Preserves fine image details	May blur edges and defects
Maintains image sharpness	Reduces image quality
Highly effective for repetitive stripe noise Ineffective against periodic interference	

Since periodic noise appears as distinct frequency peaks, it can be removed precisely without affecting important image features.

(e) Explain your answer in a logical sequence.

Step-by-Step Process

- Step 1: Capture the metal component image.
- Step 2: Electrical interference introduces periodic horizontal stripe noise.
- Step 3: Transform the image into the frequency domain using FFT.
- Step 4: Detect the periodic noise frequencies.

Step 5: Apply a notch filter to remove the unwanted frequencies.

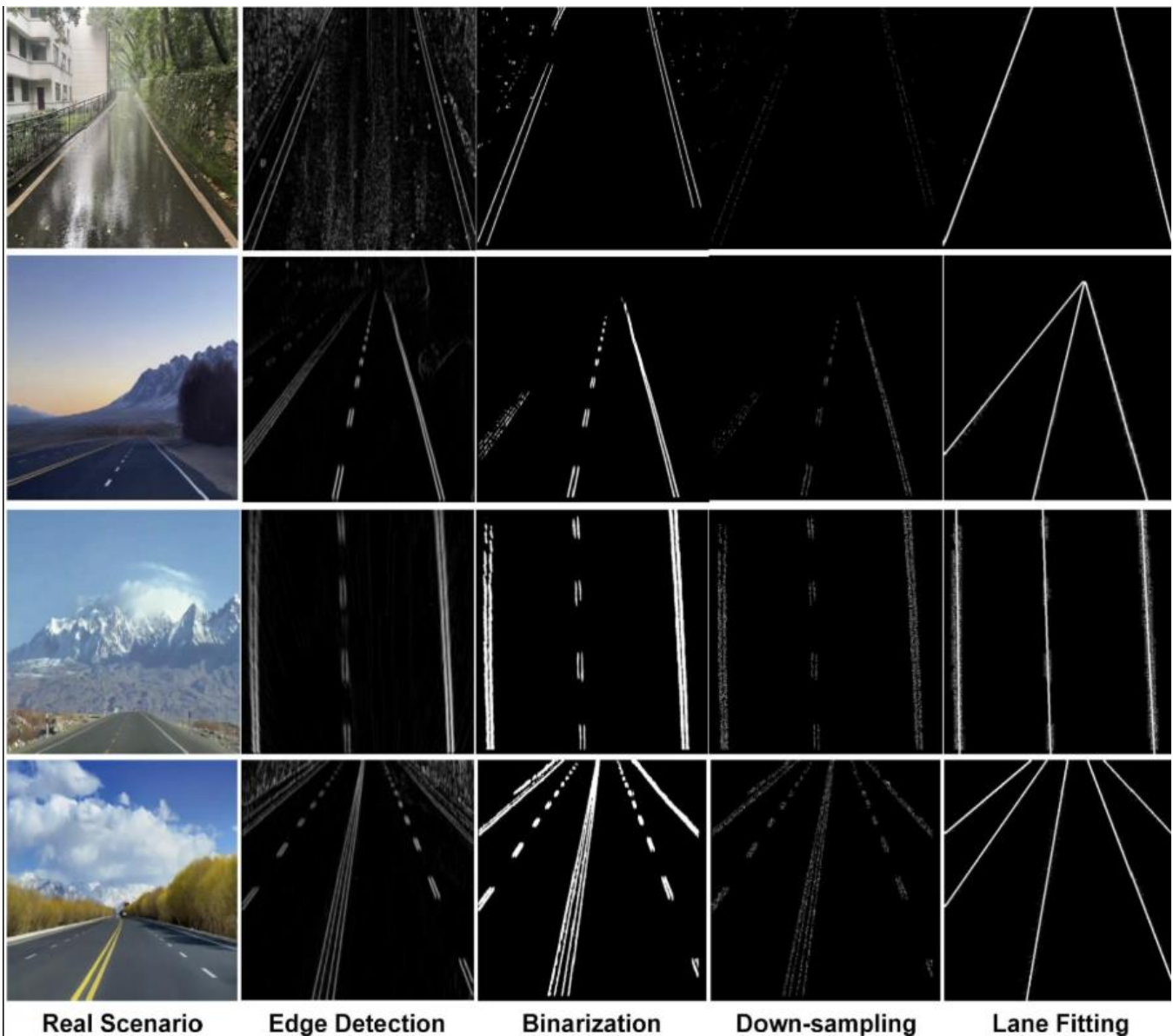
Step 6: Apply inverse FFT to reconstruct the image.

Step 7: Use the enhanced image for accurate defect detection.

Conclusion

Periodic horizontal stripe noise caused by electrical interference is best removed using a **Frequency Domain Notch Reject Filter**, as it selectively removes interference while preserving important defect information required for accurate quality inspection.

Scenario 2: Autonomous Vehicle Navigation



(a) Interpret the scenario and explain the challenge involved.

A self-driving car captures road images during heavy rain. Rain introduces image noise that reduces visibility of the road.

Challenge:

The system must remove the noise while keeping lane markings, road edges, and other important features sharp so that the vehicle can navigate safely.

(b) Identify the importance of preserving edges while removing noise.

Edge preservation is important because:

- Lane markings define the driving path.
- Road boundaries help keep the vehicle within its lane.

- Traffic signs and obstacles depend on clear edges for detection.
- Blurred edges can cause incorrect lane detection and unsafe driving decisions.

(c) Recommend a suitable filtering technique.

Recommended Technique:

Recommended filter:

- Bilateral Filter

The bilateral filter smooths noise while preserving edges by considering both:

- Spatial distance between pixels.
- Intensity (brightness) difference between pixels.

(d) Justify why your selected filter preserves important image details.

The bilateral filter preserves edges because:

- Pixels with similar intensity are averaged.
- Pixels across an edge have very different intensities and therefore receive little weight.
- Noise is reduced inside smooth regions.
- Strong edges, lane markings, and road boundaries remain sharp.

Unlike a Gaussian filter, the bilateral filter does not blur important .

(e) Present your explanation clearly and systematically.

Step-by-Step Process

Step 1: Capture the road image during rainfall.

Step 2: Rain introduces random noise into the image.

Step 3: Apply a bilateral filter.

Step 4: The filter smooths noisy regions while preserving edges.

Step 5: Detect lane markings and road boundaries.

Step 6: The autonomous vehicle uses the enhanced image for safe navigation.

Conclusion

The **bilateral filter** is the most suitable technique for this scenario because it effectively removes rain noise while preserving lane markings and road edges. This improves image quality and helps the autonomous vehicle navigate safely and accurately.

